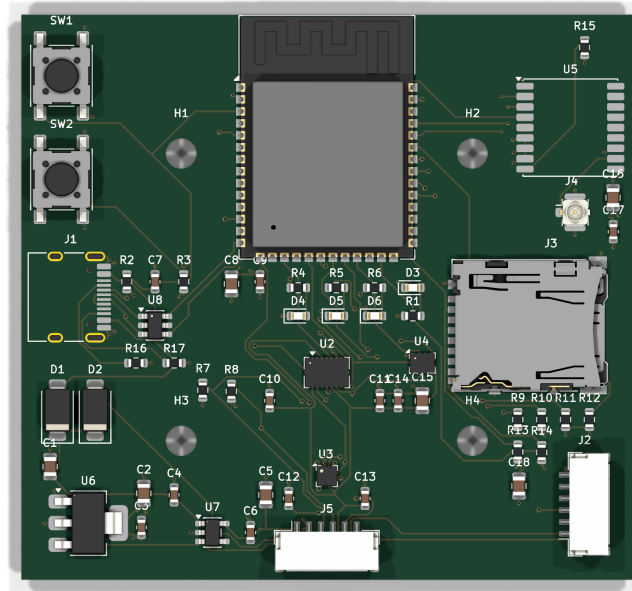


DroneMagNav rev A - assembly instructions

Magnetic-anomaly navigation module, 64 x 60 mm, 4 layers. 56 SMD parts, all on the TOP side, no through-hole parts. Files: Gerber ZIP (with FABRICATION-NOTES.md), BOM (PCBWay template), centroid file (mm, origin bottom-left, pad-1 position). This document: orientation and handling rules, IC programming, functional test, and the top assembly drawing (appendix).



Top side, as assembled (3-D render from the CAD data).

1. Orientation and polarity

Rotations in the centroid file follow the KiCad / Altium convention (degrees, counter-clockwise positive, 0 deg = library orientation). Pad-1 coordinates are given for every part; the F.Fab layer in the appendix shows pin-1 marks and pad outlines with numbers.

Ref	Part	Orientation / polarity
U1	ESP32-S3-WROOM-1-N16R8	Pin 1 = pad at the corner marked on F.Fab. The PCB antenna end of the module overhangs the TOP board edge (x 106-154 mm keep-out): do not clamp, solder or fixture that end. MSL 3: bake 24 h / 125 C if the bag was open > 168 h.
U2	BMI088 LGA-16, 0.5 mm pitch	Pin-1 dot on the package corner marked on F.Fab. Pads under the body: solder-paste stencil 0.1 mm, X-ray or AOI check recommended. Pads 2 and 4 contain filled-and-capped vias.
U3	LIS3MDL LGA-12 (magnetometer)	Pin-1 mark on F.Fab. NO magnetised tools, nozzles or trays near this part; keep it away from magnets during and after assembly.
U4	BMP280 LGA-8	Pin-1 mark on F.Fab. Do not cover the vent hole with flux residue.
U5	MAX-M10S LCC-18	Pin 1 marked on F.Fab (chamfered corner of the module). RF pin 11 faces U.FL J4.
U6	AMS1117-3.3 SOT-223	Tab = pin 2 (+3V3 output).
U7 / U8	SOT-23-5 / SOT-23-6	Pin 1 per F.Fab mark.
D1, D2	SS14 SMA Schottky	Cathode band = pad 1 (silkscreen bar).
D3-D6	0603 LEDs: D3 green, D4 green, D5 blue, D6 yellow	Cathode = pad 1 (silkscreen arrow / bar on the cathode side).
J1	USB-C HRO TYPE-C-31-M-12	Keyed by the shell; opening faces the LEFT board edge.
J2, J5	JST-GH SM06B-GHS-TB, 6-pin	Pin 1 marked on silkscreen; openings face the board edge.
J3	microSD Hirose DM3AT	Card slot faces the RIGHT board edge; check shield pads for bridges.
J4	U.FL-R-SMT-1	Centre pin = signal, keep pad 1 free of flux residue.

Ref	Part	Orientation / polarity
SW1, SW2	Tactile 6 x 6 mm	Symmetric, no polarity.
R*, C*	0603 / 0805	No polarity.

2. Process and handling

- Lead-free reflow per the sensor datasheets: peak 245-260 C, max 30 s above 217 C. No wave, no hand-soldering of U2/U3/U4.
- No ultrasonic cleaning (MEMS sensors U2, U3, U4). No-clean flux preferred.
- No magnetised tooling near U3 (magnetometer). Do not store finished boards on magnetic trays.
- PCB: 4-layer, ENIG, 1.6 mm, vias filled and capped (three vias sit inside SMD pads: U2 pads 2/4, J3 pad 4). Board edge clearance 0.5 mm; four 3.2 mm NPTH mounting holes.
- Nothing may be placed, printed or glued in the antenna keep-out at the top edge.
- Inspect J1 (USB-C) and J3 (microSD) shield pads and the U1 module pads for bridges; the U2 LGA joints by X-ray if available.

3. IC programming

No pre-programming of any part is required before assembly: the ESP32-S3 module boots into its factory ROM loader and is flashed over the USB-C connector after assembly. If the programming service is ordered, flash the supplied image **dronemagnav-firmware.factory.bin** (bootloader + partition table + application, 1.16 MB) as follows:

Step	Action
1	Connect the board to a PC with a USB-C data cable (5 V from USB is enough; no other supply).
2	Hold BOOT (SW2), press and release RESET (SW1), release BOOT. The board enumerates as a USB serial port (Espressif USB JTAG/serial debug unit).
3	With Python 3 and <i>pip install esptool</i> (any version ≥ 4), run the command below with the port name of that device.
4	Press RESET. The green PWR LED (D3) is on; the serial port re-enumerates and prints the boot log (see section 4). Done.

```
esptool --chip esp32s3 --port COM5 --baud 921600 write-flash 0x0
dronemagnav-firmware.factory.bin
```

SHA-256 of the image is listed in `firmware/release/` in the repository. The image does not depend on the microSD card; the card only carries map data.

4. Functional test (after programming)

Equipment: USB-C cable, PC with a serial terminal (115200 baud), multimeter, a phone or laptop with WiFi, optionally a FAT32 microSD card and a passive GNSS antenna with U.FL.

#	Test	Pass criteria
T1	Visual / AOI	No bridges, all 56 parts present and oriented per section 1.
T2	Power-up on USB 5 V	Green PWR LED D3 on. Supply current 60-250 mA (WiFi access point running). FAIL if < 20 mA or > 400 mA.
T3	Rails	+3V3 at U6 pin 2 (tab): 3.30 ± 0.10 V. +3.3VA at U7 pin 5: 3.30 ± 0.10 V.
T4	USB enumeration	A USB serial device appears on the PC within 3 s of RESET.
T5	Boot log (serial 115200)	Lines [imu] ok , [mag] ok , [baro] ok must appear. [sd] ok when a FAT32 card is inserted (without a card [sd] FAIL is expected and acceptable). [web] AP DroneMagNav must appear.
T6	WiFi	SSID DroneMagNav visible; password magnav123 ; <code>http://192.168.4.1/</code> loads the map page; <code>http://192.168.4.1/api/status</code> returns JSON with non-zero magnetic field and pressure values.
T7	Buttons	RESET repeats the boot log. BOOT held during RESET stops the boot log (download mode) - then RESET again to resume.
T8	GNSS (optional)	Antenna on J4, board near a window: blue FIX LED D5 on within 3 min; <code>/api/status</code> shows a latitude/longitude.
T9	LEDs D4 NAV and D6 LINK	Off during this test (they need a flight controller / navigation mode). They must not be stuck on.

Record per board: serial port seen, T2 current, T3 voltages, T5 log lines, T6 result. A board passing T1-T7 is a good unit.